

REVIEW EX

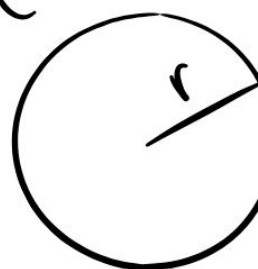
By Sir Adnan Ullah Khan

2

$$\frac{b}{c} = 1$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$y - y_1 = m(x - x_1)$$



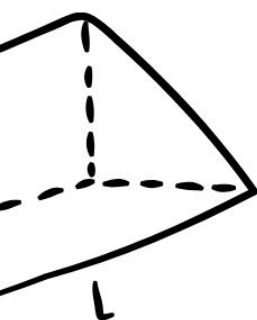
$$C = 2\pi r$$

$$S = \frac{d}{t}$$

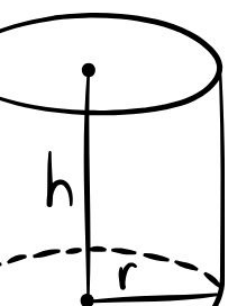
$$\frac{+}{-} = -$$

$$\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\frac{V_f - V_i}{+}$$

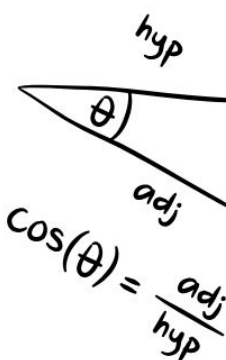


bhl

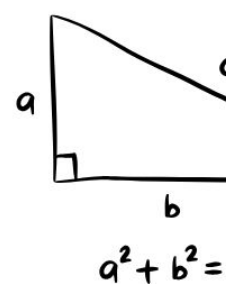


$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\frac{\sqrt{b^2 - 4ac}}{2a}$$



$$\cos(\theta) = \frac{\text{adj}}{\text{hyp}}$$



$$a^2 + b^2 = c^2$$

(Q1)

For $|A_y|$:

$$|A_y| = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$$

$$|A_y| = \begin{vmatrix} 1 & -4 & 0 \\ 3 & -5 & 0 \\ 2 & -1 & 1 \end{vmatrix}$$

$$|A_y| = 1(-5 \cdot 0 + 4 \cdot 3 \cdot 0) + 0(3 \cdot -1 - 2 \cdot 3) + 0(2 \cdot -5 - 3 \cdot -1)$$

$$|A_y| = 1(-5 \cdot 0) + 4(3 \cdot 0) + 0(-3 + 10)$$

$$|A_y| = -5 + 12$$

$$|A_y| = 7$$

For $|A_z|$:

$$|A_z| = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

$$|A_z| = \begin{vmatrix} 1 & -2 & -4 \\ 3 & 1 & -5 \\ 2 & 0 & -1 \end{vmatrix}$$

$$|A_z| = 1(1 \cdot -5 + 2 \cdot 3 \cdot -5) - 4(3 \cdot -1 - 2 \cdot 2)$$

$$|A_z| = 1(-5 - 10) - 4(-3 + 10)$$

$$|A_z| = -15 - 28$$

$$|A_z| = -43$$

By applying Cramer's Rule we get:

$$\Rightarrow x = \frac{|A_x|}{|A|}$$

$$\Rightarrow x = \frac{-14}{7}$$

$$\Rightarrow x = -2$$

$$\Rightarrow y = \frac{|A_y|}{|A|}$$

$$\Rightarrow y = \frac{7}{7}$$

$$\Rightarrow y = 1$$

$$\Rightarrow z = \frac{|A_z|}{|A|}$$

$$\Rightarrow z = \frac{21}{7}$$

$$\Rightarrow z = 3$$

Thus,

 $x = -2, y = 1$ and $z = 3$ is the required solution.

Review Exercise

(100)

Q1) Find the value of x and y if: $\begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 8 & 14 \end{bmatrix}$

Sol:

$$= \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 8 & 14 \end{bmatrix}$$

$$= \begin{bmatrix} x+3 & 5-4 \\ 7+1 & y-3+2 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 8 & 14 \end{bmatrix}$$

$$= \begin{bmatrix} x+3 & 1 \\ 8 & y-1 \end{bmatrix} = \begin{bmatrix} 7 & 1 \\ 8 & 14 \end{bmatrix}$$

Now equating,

$$\Rightarrow x+3 = 7 \text{ --- (1)}$$

$$\Rightarrow y-1 = 14 \text{ --- (2)}$$

Take (1)

$$\Rightarrow x+3 = 7$$

$$\Rightarrow x = 4$$

Take (2)

$$\Rightarrow y-1 = 14$$

$$\Rightarrow y = 14+1$$

$$\Rightarrow y = 15$$

Thus,

$$x = 4 \text{ and } y = 15$$

Q2) Calculate AC , BC and $(A+B)C$. Also verify that $(A+B)C =$

for:

$$A = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 0 & 2 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$$

Sol:

AC:

$$\Rightarrow AC = \begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{3 \times 1}$$

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$$\Rightarrow AC = \begin{bmatrix} (0)(2) + (4)(-2) + (7)(3) \\ (-5)(2) + (0)(-2) + (9)(3) \\ (1)(2) + (-7)(-2) + (0)(3) \end{bmatrix}_{3 \times 1}$$

$$\Rightarrow AC = \begin{bmatrix} 9 \\ 12 \\ 30 \end{bmatrix}_{3 \times 1}$$

$$BC = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{2 \times 1}$$

$$\Rightarrow BC = \begin{bmatrix} (0)(2) + (1)(-2) + (1)(3) \\ (1)(2) + (-2)(0) + (2)(3) \\ (1)(2) + (2)(-2) + (0)(3) \end{bmatrix}_{3 \times 1}$$

$$\Rightarrow BC = \begin{bmatrix} 1 \\ 8 \\ -2 \end{bmatrix}_{3 \times 1}$$

(10)

(A+B)C:

First we find A+B.

$$\Rightarrow A+B = \begin{bmatrix} 0 & 6 & 7 \\ -5 & 0 & 9 \\ 7 & -8 & 0 \end{bmatrix}_{3 \times 3} + \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A+B = \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix}_{3 \times 3}$$

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Now,

$$= (A+B)C$$

$$= \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} (0)(2) + (7)(-2) + (8)(3) \\ (-5)(2) + (0)(-2) + (10)(3) \\ (8)(2) + (-6)(-2) + (0)(3) \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 10 \\ 20 \\ 28 \end{bmatrix}_{3 \times 1}$$

To Prove:

$$(A+B)C = AC + BC$$

So,

Take L.H.S

$$= (A+B)C$$

$$\Rightarrow A+B = \begin{bmatrix} 0 & 6 & 7 \\ -5 & 0 & 9 \\ 7 & -8 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}_{3 \times 3} = \begin{bmatrix} 9 \\ 12 \\ 30 \end{bmatrix}_{3 \times 1}$$

$$\Rightarrow A+B = \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix}_{3 \times 3}$$

Now,

$$= (A+B)C$$

$$= \begin{bmatrix} 0 & 7 & 8 \\ -5 & 0 & 10 \\ 8 & -6 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} (0)(2) + (7)(-2) + (8)(3) \\ (-5)(2) + (0)(-2) + (10)(3) \\ (8)(2) + (-6)(-2) + (0)(3) \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 10 \\ 20 \\ 28 \end{bmatrix}_{3 \times 1}$$

Take R.H.S

$$= AC + BC$$

First we find AC:

$$= \begin{bmatrix} 0 & 6 & 7 \\ -5 & 0 & 9 \\ 7 & -8 & 0 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} (0)(2) + (6)(-2) + (7)(3) \\ (-5)(2) + (0)(-2) + (9)(3) \\ (7)(2) + (-8)(-2) + (0)(3) \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 9 \\ 12 \\ 30 \end{bmatrix}_{3 \times 1}$$

Now we find BC:

$$= BC = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}_{2 \times 1}$$

(103)

$$= \begin{bmatrix} (1)(2) + (1)(-3) + (1)(3) \\ (1)(2) + (0)(-3) + (1)(3) \\ (1)(2) + (1)(-3) + (1)(3) \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}_{3 \times 1}$$

$$\text{Now, } AC + BC$$

$$= \begin{bmatrix} 9 \\ 12 \\ 20 \end{bmatrix}_{3 \times 1} + \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}_{3 \times 1}$$

$$= \begin{bmatrix} 10 \\ 20 \\ 28 \end{bmatrix}_{3 \times 1}$$

$\Rightarrow$ L.H.S = R.H.S proved

Q If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ show that $A^2 - 5A + 7I = 0$

Taking L.H.S

$$= A^2 - 5A + 7I$$

First we find A^2

$$= A^2$$

$$= \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} (3)(3) + (1)(-1) & (3)(1) + (1)(2) \\ (-1)(3) + (2)(-1) & (-1)(1) + (2)(2) \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}_{2 \times 2}$$

Now,

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}_{2 \times 2} + 5 \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}_{2 \times 2} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 23 & 10 \\ -10 & 13 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}_{2 \times 2}$$

$$= \begin{bmatrix} 30 & 10 \\ -10 & 20 \end{bmatrix}_{2 \times 2}$$

(104)

D Find the degree or index of the nilpotent matrix

$$\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3} \text{ Sol}$$

$$\text{Let, } A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3}$$

Now,

$$\Rightarrow A^2 = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (1)(1) + (-3)(-1) + (-4)(1) & (1)(-3) + (-3)(3) + (-4)(-1) & (1)(-4) + (-3)(4) + (-4)(-1) \\ (-1)(1) + (3)(-1) + (4)(1) & (-1)(-3) + (3)(3) + (4)(-1) & (-1)(-4) + (3)(4) + (4)(-1) \\ (1)(1) + (-3)(-1) + (-4)(1) & (1)(-3) + (-3)(3) + (-4)(-1) & (1)(-4) + (-3)(4) + (-4)(-1) \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+3-4 & -3-9+4 & -4-12+4 \\ -1-3+4 & 3+9-4 & 4+12-4 \\ 1+3-4 & -3-9+4 & -4-12+4 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 0$$

$\therefore A^2 = 0$, so A is nilpotent matrix of index 2.

Q Show that $\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is periodic matrix of period 2.

Let,

$$A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

(105)

Now,

$$\Rightarrow A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (1)(1)+(-2)(-3)+(-6)(2) & (1)(-2)+(-2)(2)+(9)(0) & (1)(-6)+(-2)(9)+(-6)(-3) \\ (-3)(1)+(2)(-3)+(9)(2) & (-3)(-2)+(2)(2)+(9)(0) & (-3)(-6)+(2)(9)+(9)(-3) \\ (2)(1)+(0)(-3)+(-3)(2) & (2)(-2)+(0)(2)+(-3)(0) & (2)(-6)+(0)(9)+(-3)(-3) \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+6-12 & -2-4+0 & -6-18+18 \\ -3-6+18 & 6+4+0 & 18+18-27 \\ 2+0-6 & -4+0+0 & -12+0+9 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}_{3 \times 3}$$

Now we again multiply by A.

$$\Rightarrow A^3 = A^2 \cdot A$$

$$\Rightarrow A^3 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^3 = \begin{bmatrix} (-5)(1)+(-6)(-3)+(-6)(2) & (-5)(-2)+(-6)(2)+(-6)(0) & (-5)(-6)+(-6)(9)+(-6)(-3) \\ (9)(1)+(-6)(-3)+(9)(2) & (9)(-2)+(-6)(2)+(9)(0) & (9)(-6)+(-6)(9)+(9)(-3) \\ (-4)(1)+(-4)(-3)+(-3)(2) & (-4)(-2)+(-4)(2)+(-3)(0) & (-4)(-6)+(-4)(9)+(-3)(-3) \end{bmatrix}$$

$$\Rightarrow A^3 = \begin{bmatrix} -5+18-12 & 10-12+0 & 30-54+18 \\ 9-30+18 & -18+20+0 & -54+90-27 \\ -4+12-6 & 8-8+0 & 24-36+9 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^3 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}_{3 \times 3}$$

(106)

 $\therefore A^3 = A$ so, A is a periodic matrix of period 2.

Which of the following is idempotent or involutory.

$$(i) \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol

$$\text{Let } A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

Now,

$$\Rightarrow A^2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (1)(1)+(-1)(0)+(0)(0) & (1)(-1)+(-1)(-1)+(0)(0) & (1)(0)+(-1)(0)+(0)(1) \\ (0)(1)+(-1)(0)+(0)(0) & (0)(-1)+(-1)(-1)+(0)(0) & (0)(0)+(-1)(0)+(0)(1) \\ (0)(1)+(0)(0)+(1)(0) & (0)(-1)+(0)(-1)+(1)(0) & (0)(0)+(0)(0)+(1)(1) \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1+0+0 & -1+1+0 & 0+0+0 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

 $\therefore A^2 = I$. So A is an involutory matrix.

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(ii)
$$\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$$

Sol

let, $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}_{3 \times 3}$

Now,

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}_{3 \times 3} \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} (0)(0) + (1)(4) + (-1)(3) & (0)(1) + (1)(-3) + (-1)(-3) & (0)(-1) + (1)(4) + (-1)(4) \\ (4)(0) + (-3)(4) + (4)(3) & (4)(1) + (-3)(-3) + (4)(-3) & (4)(-1) + (-3)(4) + (4)(4) \\ (3)(0) + (-3)(4) + (4)(3) & (3)(1) + (-3)(-3) + (4)(-3) & (3)(-1) + (-3)(4) + (4)(4) \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ -12+12 & 3+9-12 & -3-12+16 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

$\therefore A^2 = I$. So A is an involutory matrix.

(107)

Q) Which of the following are Hermitian or skew-Hermitian.

01)
$$\begin{bmatrix} 1 & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix}$$

Sol

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let, $A = \begin{bmatrix} 1 & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix}_{3 \times 3}$

Now

$$\Rightarrow \bar{A} = \begin{bmatrix} 1 & 1+i & 2 \\ 1+i & -3i & -i \\ -2 & -i & 0 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow (\bar{A})^t = \begin{bmatrix} 1 & 1+i & 2 \\ 1+i & -3i & -i \\ 2 & -i & 0 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow (\bar{A})^t = - \begin{bmatrix} -1 & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix}_{3 \times 3}$$

$\therefore (\bar{A})^t$ is neither equal to A nor equal to $-A$. So it is neither a hermitian matrix nor skew-hermitian.

(108)
$$\begin{bmatrix} 4 & 1-i & 7 \\ 1+i & 6 & -i \\ 7 & i & 5 \end{bmatrix}$$

Sol

let, $A = \begin{bmatrix} 4 & 1-i & 7 \\ 1+i & 6 & -i \\ 7 & i & 5 \end{bmatrix}_{3 \times 3}$

Now,

$$\Rightarrow \bar{A} = \begin{bmatrix} 4 & 1+i & 7 \\ 1-i & 6 & i \\ -7 & -i & 5 \end{bmatrix}_{3 \times 3}$$

$$\Rightarrow (\bar{A})^t = \begin{bmatrix} 4 & 1-i & 7 \\ 1+i & 6 & -i \\ 7 & i & 5 \end{bmatrix}_{3 \times 3}$$

$\therefore (\bar{A})^t = A$
 $\therefore A$ is Hermitian Matrix.

11) Show that $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = 0$

Take L.H.S

$$= \begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$$

Take 2 as common from Row 2

$$= 2 \begin{vmatrix} a & b & c \\ a+x & b+y & c+z \\ x & y & z \end{vmatrix}$$

$$= 2 \begin{vmatrix} a+x & b+y & c+z \\ a+x & b+y & c+z \\ x & y & z \end{vmatrix} \quad R_1 + R_3$$

$\therefore$ Two rows are identical
 $\therefore$ determinant is equal to zero.

$$= 2(0)$$

$$= 0$$

$$= R.H.S$$

(67)

2

2

f

1

12) Solve following system of linear equations:-

$$2x+4y-z=0, \quad x-2y-2z=2 \quad \text{and} \quad -5x-8y+3z=1$$

Sol

Your Success is Our Win
CRAMER'S RULE

First write system of linear equations in matrix form i.e.,

$$A \cdot X = B, \quad \text{where } A = \begin{bmatrix} 2 & 4 & -1 \\ 1 & -2 & -2 \\ -5 & -8 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Here,

$$a_1 = 2, b_1 = 4, c_1 = -1, d_1 = 0$$

$$a_2 = 1, b_2 = -2, c_2 = -2, d_2 = 2$$

$$a_3 = -5, b_3 = -8, c_3 = 3, d_3 = 1$$

For $|A|$:

$$\therefore |A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 2 & 4 & -1 \\ 1 & -2 & -2 \\ -5 & -8 & 3 \end{vmatrix}$$

$$\Rightarrow |A| = 2 \begin{vmatrix} -2 & -2 \\ -8 & 3 \end{vmatrix} - 4 \begin{vmatrix} 1 & -2 \\ -5 & 3 \end{vmatrix} - 1 \begin{vmatrix} 1 & -2 \\ -5 & -8 \end{vmatrix}$$

$$\Rightarrow |A| = 2(-6-16) - 4(3-10) - 1(-8-10)$$

$$\Rightarrow |A| = 2(-22) - 4(-7) - 1(-18)$$

$$\Rightarrow |A| = -44 + 28 + 18$$

$$\Rightarrow |A| = 2$$

For $|A_x|$:

$$\therefore |A_x| = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$$

First we find A^{-1} :

$$|A| = \begin{vmatrix} 2 & 4 & -1 \\ 1 & -2 & -2 \\ -5 & -8 & 3 \end{vmatrix}$$

$$\Rightarrow |A| = 2 \begin{vmatrix} -2 & -2 & -4 \\ -8 & 3 & 3 \end{vmatrix} - 4 \begin{vmatrix} 1 & -2 & -1 \\ -5 & 3 & -5 \end{vmatrix} - 1 \begin{vmatrix} 1 & -2 \\ -5 & -8 \end{vmatrix}$$

$$\Rightarrow |A| = 2(-6-16) - 4(3-10) - 1(-8-10)$$

$$\Rightarrow |A| = -44 + 46$$

$$\Rightarrow |A| = 2$$

$$\text{adj of } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t$$

Co-factors of A:

$$\therefore A_{ij} = (-1)^{i+j} M_{ij}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} -2 & -2 \\ -8 & 3 \end{vmatrix}$$

$$A_{13} = (-1)^{1+3} (-8-10)$$

$$A_{13} = -18$$

$$A_{12} = (-1)^{1+2} (-6-16)$$

$$A_{12} = -22$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 4 & -1 \\ -8 & 3 \end{vmatrix}$$

$$A_{21} = (-1)^3 (12-8)$$

$$A_{21} = -4$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & -2 \\ -5 & 3 \end{vmatrix}$$

$$A_{12} = (-1)^{1+2} (3-10)$$

$$A_{12} = 7$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & -1 \\ -5 & 3 \end{vmatrix}$$

$$A_{22} = (-1)^{2+2} (6-5)$$

$$A_{22} = 1$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1 & -2 \\ -5 & -8 \end{vmatrix}$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 4 \\ -5 & -8 \end{vmatrix}$$

$$A_{23} = (-1)^5 (-16+20)$$

$$A_{23} = -4$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} 4 & -1 \\ -2 & -2 \end{vmatrix}$$

$$A_{31} = (-1)^4 (-8-2)$$

$$A_{31} = -10$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix}$$

$$A_{32} = (-1)^5 (-4+1)$$

$$A_{32} = -3$$

$$A_{33} = (-1)^{3+3} \begin{vmatrix} 2 & 4 \\ 1 & -2 \end{vmatrix}$$

$$A_{33} = (-1)^4 (-4-4)$$

$$A_{33} = -8$$

$$\text{adj of } A = \begin{bmatrix} -22 & 7 & -18 \\ -4 & 1 & -4 \\ -10 & -3 & -8 \end{bmatrix}^t$$

$$\text{adj of } A = \begin{bmatrix} -22 & -4 & -10 \\ 7 & 1 & -3 \\ -18 & -4 & -8 \end{bmatrix}$$

$A^{-1} = \frac{\text{adj of } A}{|A|}$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} -22 & -4 & -10 \\ 7 & 1 & -3 \\ -18 & -4 & -8 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -11 & -2 & -5 \\ 7/2 & 1/2 & 3/2 \\ -9 & -2 & -4 \end{bmatrix}$$

$\therefore X = A^{-1}B$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -11 & -2 & -5 \\ 7/2 & 1/2 & 3/2 \\ -9 & -2 & -4 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (-11)(0) + (-2)(2) + (-5)(-2) \\ (7/2)(0) + (1/2)(2) + (3/2)(-2) \\ (-9)(0) + (-2)(2) + (-4)(-2) \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4+10 \\ 1-3 \\ -4+8 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ -2 \\ 4 \end{bmatrix}$$

Thus $x=6, y=-2, z=4$

$$\text{S-set} = \left\{ (6, -2, 4) \right\}$$

Gauss Elimination Method

First, write given system of linear equations in matrix form i.e. $AX=B$

Here,

$$A = \begin{bmatrix} 2 & 4 & -1 \\ 1 & -2 & -2 \\ -5 & -7 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$$

Now we form augmented matrix (A_B) by including B into B-

$$A_B = \begin{bmatrix} 2 & 4 & -1 & 0 \\ 1 & -2 & -2 & 2 \\ -5 & -8 & 3 & -2 \end{bmatrix}$$

Reduce it to echelon form

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 2 & 4 & -1 & 0 \\ -5 & -8 & 3 & -2 \end{bmatrix} R_{12}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 2-2 & 4+4 & -1+4 & 0-4 \\ -5+5 & -8-10 & 3-10 & -2+10 \end{bmatrix} \begin{matrix} \\ R_2 + (-2R_1) \\ R_3 + (5R_1) \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 8 & 3 & -4 \\ 0 & -18 & -7 & 8 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & -18 & -7 & 8 \end{bmatrix} (1/8)R_2$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & -18+18 & -7+18(3/8) & 8+18(-1/2) \end{bmatrix} R_3 + 18R_2$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & 0 & -1/4 & -1 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & 0 & 1 & 4 \end{bmatrix} (-4)R_3$$

From above matrix we get an equivalent matrix equation.

$$x + 2y - 2z = 2 \quad \text{--- (1)}$$

$$y + (3/8)z = -1/2 \quad \text{--- (2)}$$

$$z = 4 \quad \text{--- (3)}$$

Put z in (2)

$$y + \frac{3}{8}(4) = -\frac{1}{2}$$

$$y = \frac{-1 - \frac{3}{2}}{2}$$

$$y = \frac{-4}{2}$$

$$y = -2$$

$$x - 4 = 2$$

$$x = 2 + 4$$

$$x = 6$$

Thus, $x = 6, y = -2$ & $z = 4$

$$\therefore \text{Set} = \{(6, -2, 4)\}$$

Put y & z in (1)

$$x - 2(-2) - 2(4) = 2$$

$$x + 4 - 8 = 2$$

Gauss Jordan Method

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First, write given system of linear equations in matrix form i.e. $AX=B$

Here,

$$A = \begin{bmatrix} 2 & 4 & -1 \\ 1 & -2 & -2 \\ -5 & -8 & 3 \end{bmatrix}_{3 \times 3}, B = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}_{3 \times 1} \text{ and } X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3 \times 1}$$

Now,

we form augmented matrix by including B into A.

$$A_B = \begin{bmatrix} 2 & 4 & -1 & 0 \\ 1 & -2 & -2 & 2 \\ -5 & -8 & 3 & -2 \end{bmatrix}_{3 \times 4}$$

Reduced it into reduced echelon form.

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 2 & 4 & -1 & 0 \\ -5 & -8 & 3 & -2 \end{bmatrix} R_{12}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 2-2 & 4+4 & -1+4 & 0-4 \\ -5+5 & -8-10 & 3-10 & -2+10 \end{bmatrix} \begin{matrix} \\ R_2 + (-2)R_1 \\ R_3 + (5)R_1 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 8 & 3 & -4 \\ 0 & -18 & -7 & 8 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & -18 & -7 & 8 \end{bmatrix} (1/8)R_2$$

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$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & -18 & -7 & 8 \end{bmatrix} \begin{matrix} \\ R_3 + 18R_2 \end{matrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & 0 & -1/4 & -1 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 & -1/2 \\ 0 & 0 & 1 & 4 \end{bmatrix} (-4)R_3$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 3/8 - 1/8 & -1/2 - 4(1/8) \\ 0 & 0 & 1 & 4 \end{bmatrix} R_2 + (-3/8)R_3$$

$$A_B \sim \begin{bmatrix} 1 & -2 & -2 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix}$$

$$A_B \sim \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix} R_1 + 2R_2$$

$$A_B \sim \begin{bmatrix} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{bmatrix} R_1 + 2R_3$$

From above matrix we get an equivalent matrix eq.

$$x + 0y + 0z = 6$$

$$0x + y + 0z = -2$$

$$0x + 0y + z = 4$$

Thus,

$$x=6, y=-2 \text{ \& } z=4$$

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Q Show that
$$\begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix} = 2(x+y+z)^3$$

Sol

Take L.H.S

$$= \begin{vmatrix} x+y+2z & x & y \\ z & y+z+2x & y \\ z & x & z+x+2y \end{vmatrix}$$

$$= \begin{vmatrix} x+y+2z & x+y & y \\ z & y+z+2x & y \\ z & z+2x+2y & z+x+2y \end{vmatrix} \quad C_2 + C_3 \rightarrow C_2$$

$$= \begin{vmatrix} 2x+2y+2z & x+y & y \\ 2x+2y+2z & 2y+z+2x & y \\ 2x+2y+2z & z+2x+2y & z+x+2y \end{vmatrix} \quad C_1 + C_2$$

Taking $2x+2y+2z$ as common from Column 1

$$= 2x+2y+2z \begin{vmatrix} 1 & x+y & y \\ 1 & 2y+z+2x & y \\ 1 & z+2x+2y & z+x+2y \end{vmatrix}$$

Expand by Row 1.

$$= 2(x+y+z) \left\{ \begin{vmatrix} 2y+z+2x & y \\ z+2x+2y & z+x+2y \end{vmatrix} - (x+y) \begin{vmatrix} 1 & y \\ 1 & z+x+2y \end{vmatrix} + y \begin{vmatrix} 1 & 2y+z+2x \\ 1 & z+2x+2y \end{vmatrix} \right\}$$

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$$= 2(x+y+z) \left[1 \left\{ (2y+z+2x)(z+x+2y) - (y)(z+2x+2y) \right\} - (x+y)(z+2x+2y) + y(z+2x+2y - z - x - 2y) \right]$$

$$= 2(x+y+z) \left[1 \left\{ (2yz+2xy+4y^2+z^2+xz+2yz+2xz+2x^2+4xy) - (yz+2xy+2y^2) \right\} - (x+y)(z+x+y) + y(0) \right]$$

$$= 2(x+y+z) \left[\left\{ (2yz+2xy+4y^2+z^2+xz+2yz+2xz+2x^2+4xy - yz - 2xy - 2y^2) \right\} - (xz+x^2+xy+yz+xy+y^2) \right]$$

$$= 2(x+y+z) \left[(2yz+2yz-yz+2xy+4xy-2xy+4y^2-2y^2+z^2+xz+2xz+2x^2+2x^2) - xz-x^2-xy-yz-xy-y^2 \right]$$

$$= 2(x+y+z) (3yz+4xy+2y^2+z^2+3xz+2x^2-xz-x^2-xy-yz-xy-y^2)$$

$$= 2(x+y+z) (3yz+4xy+2y^2+z^2+3xz+2x^2-xz-x^2-xy-yz-xy-y^2)$$

$$= 2(x+y+z) (2yz+2xy+y^2+z^2+2xz+x^2)$$

$$= 2(x+y+z) (x^2+y^2+z^2+2xy+2yz+2zx)$$

$$\therefore a^2+b^2+c^2+2ab+2bc+2ca = (a+b+c)^2$$

$$= 2(x+y+z)(x+y+z)^2$$

$$= 2(x+y+z)^3$$

= R.H.S proved!

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) =$$

$$\begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{a} & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & \frac{1}{b} & \frac{1}{b} \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

$$abc \left(\frac{1}{a} + \frac{1}{b} + 1 + \frac{1}{c} \right)$$

$$\textcircled{ii} \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \quad R_1 \rightarrow R_1 - R_2 - R_3$$

$$\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} a & a \\ c+a & b \\ c & a+b \end{vmatrix}$$

$$\begin{vmatrix} c & b \\ c+a & b \\ c & a+b \end{vmatrix}$$

$$\begin{vmatrix} c & b \\ c+a & b \\ c & a+b \end{vmatrix}$$

$$\begin{vmatrix} c & b \\ c+a & b \\ c & a+b \end{vmatrix}$$

$$\begin{vmatrix} c & b \\ c+a & b \\ c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

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$$\begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$